

Homework #3

Illya Starikov

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Problem #1

- (a) Illegal (*ADD only applies to A.*)
- (b) Illegal (*255H is too large*)
- (c) Legal
- (d) Illegal (*Same reasoning as (a)*)
- (e) Legal
- (f) Illegal (*F5H will be treated as a label instead of a number.*)

Problem #2

```
MOV R4, #25H      ; Moves 37 into register 4.
MOV A, #1FH       ; Moves 31 into temporary register A
ADD A, R4         ; Adds register 4's value to register A's value
```

It's a simple add operation, resulting in the value of $37_{10} + 31_{10} = 68_{10} = 44_{16}$; it's kept in ROM.

Problem #3

```

ORG 0;
MOV R4, #0FH      ; Move 0F to register 4
MOV A, #0A2H      ; Move A2 to register A
ADD A, R4         ; Add R4 To A
MOV R7, A         ; Move the result to R7
END;

```

Address	Contents
0000	0F
0001	A2
0002	B1
0003	B1

Problem #4

```

ORG 0

MOV R0, #30H      ; Move the value 30H to R0
LOOP: MOV @R0, #0FFH ; Move FFH at R0
      INC R0       ; Increment R0
      CJNE R0, #40H, LOOP
      END

```

Problem #5

```

ORG 0;

SETB PSW.3      ; Switch to register bank 3
MOV R0, #0A5H   ; Load A5 into register 0
PUSH 0          ; Push R0 onto stack

SETB PSW.0      ; Switch to register bank 4
POP 5           ; Pop onto register 5

END;

```

Problem #6

Assuming A is the register to store results,

```

ORG 0;

```

```

MOV R6, #5FH          ; Load 5F into R6
MOV R5, #01001001B    ; Load 01001001 into R5

/* Swap */
MOV A, R6;             ; temp (A) = R6
MOV 6, 5;              ; R6 = R5
MOV R5, A              ; R5 = Temp (A)

/* Add */
ADD A, R6;             ; A already had R6's values, just add original R5
                     ; Which got switched.. so actually add R6
ADD A, #20;

END;

```